Atish Kadam

cs25mtech14003@iith.ac.in | atishkadam158@gmail.com | +91 8010432751 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.16
Lord Buddha Public School Kota, Rajasthan	Class XII (CBSE)	2020	85.40%
Jawahar Navodaya Vidyalaya Pokharapur, Tal-Mohol,Dist- Solapur Maharashtra	Class X (CBSE)	2018	92.20%

EXPERIENCE

Seagate Technology (India) Private Limited | **Software Firmware Intern** |

- Developed **enclosure management firmware** for JBOD storage systems using **C, C++, and Python.**
- Engineered firmware on an **OpenBMC-based embedded platform** using **Yocto Linux and ASPEED AST2600**, enabling reliable system management.
- Implemented **thermal and power monitoring/control** and integrated **Redfish, CLI, and serial interfaces** to enhance system manageability.
- Diagnosed and resolved **20+ system-level firmware bugs** through Linux-based debugging, testing, and validation, contributing to **system release and software stability**.

MASTER'S THESIS

Learning Transferable Representations Through RL-Based Graph Optimization

- Developed an **RL-based graph structure learning framework** to discover task-relevant relationships among tabular features for downstream prediction.
- Formulated **graph construction as a policy optimization problem** and trained the graph-generation policy using **REINFORCE**, with prediction performance as the reward signal.
- Applied **GNN-based message passing and multi-scale feature fusion** to capture feature interactions and higher-order dependencies from the learned graph.
- Fused **learned graph and textual representations** to jointly capture structural and semantic information for **multimodal tabular prediction**.

PROJECTS

Multimodal Product Search Engine

- Built a multimodal search engine for 5k+ images using CLIP (ViT-B/32) and Qdrant, supporting text, image, and image-text queries.
- Combined BM25 keyword retrieval with CLIP dense search using weighted score fusion to improve product retrieval.
- Implemented Cross-Encoder reranking using Sentence-Transformers (ms-marco-MiniLM-L-6-v2) to refine search results.
- Evaluated 50 queries, achieving 0.95 MRR, 0.75 NDCG, 75% top-10 precision, and 32ms mean hybrid latency.

Layout-Aware Multimodal RAG for Structured Documents

- Built a **3-modality RAG pipeline** for financial PDFs using **Docling** to parse text, tables, and figures, preserving table structure in Markdown and generating **VLM-based captions** for visual regions.
- Engineered a **2-stage retrieval pipeline** using **384-D BGE embeddings, Qdrant vector search, and cross-encoder re-ranking**, with modality-aware routing and runtime VLM re-querying for figure-based questions.
- Developed a **2-pipeline ablation framework** comparing layout-aware and plain-text RAG across **3 QA categories** (table, figure, text), evaluated using **Exact Match, F1, and ROUGE-L**.

Geometric Object Detection via Modified ResNet-18 | CS5480 Deep Learning Hackathon

- Designed a ResNet-18 binary classifier from scratch, integrating Global Sum Pooling and a dropout-regularized MLP classification head.
- Engineered a PyTorch pipeline with image augmentation, stratified 4-fold cross-validation, mixed-precision training.

- Implemented soft-voting ensembling across four fold-trained models, achieving **0.7813 Kaggle leaderboard accuracy**.

SKILLS

- **Programming:** C, C++, Python, Bash, SQL
- **Web Development:** HTML, CSS, JavaScript
- **Core CS:** Data Structures & Algorithms, Operating Systems, Computer Networks, Database Management Systems
- **Machine Learning:** PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib

RELEVANT COURSES

- **Machine Learning:** Foundations of Machine Learning, Deep Learning, Visual Computing, Fraud Analytics
- **Networks & Security:** Advanced Computer Networks, Network Security
- **Core CS & Theory:** Advanced Data Structures & Algorithms, Linear Optimization

SCHOLASTIC ACHIEVEMENTS

- Secured Elite Topper certificate in Database Management System NPTEL Course
- Secured Elite Topper certificate in Discrete Mathematics NPTEL Course